Money Laundering Detection Using Machine Learning

Abstract

Money laundering is a major financial crime that enables illegal funds to be disguised as legitimate transactions. Financial institutions process millions of transactions daily, making manual detection extremely difficult. This project focuses on developing a Machine Learning-based Money Laundering Detection System capable of identifying suspicious financial transactions automatically.

The solution utilizes transaction-level data containing information such as transaction amount, payment currency, bank locations, payment methods, and laundering categories. Various data preprocessing techniques including timestamp engineering, categorical encoding, class balancing, and feature selection were applied. Three classification models, namely Logistic Regression, Decision Tree, and Random Forest, were trained and evaluated.

Experimental results demonstrated that the Random Forest classifier provided the highest predictive performance . The trained model was deployed through a FastAPI backend and tested through API endpoints for real-time prediction.

1. Introduction

Financial institutions process millions of transactions every day, making it increasingly difficult to manually identify suspicious activities associated with money laundering. Criminals often attempt to disguise illegally obtained funds by transferring money through multiple accounts, currencies, and financial channels, making detection a complex task. Traditional rule-based monitoring systems generate a large number of alerts and often fail to identify sophisticated laundering patterns, resulting in false positives and delayed investigations.

With the advancement of Machine Learning, financial organizations can now analyze large transaction datasets and automatically discover hidden patterns that indicate potential money laundering activities. Machine Learning models can learn from historical transaction data and classify new transactions as either legitimate or suspicious based on learned behavior.

This project focuses on developing a Money Laundering Detection System using Machine Learning techniques. The solution analyzes transaction-related attributes such as transaction amount, payment currency, bank locations, payment methods, timestamps, and laundering categories to predict whether a transaction is involved in money laundering. The system aims to improve detection accuracy, reduce manual effort, and support financial institutions in combating financial crime.

2. Problem Statement

Money laundering is one of the most significant challenges faced by banks, financial institutions, and regulatory authorities. As digital banking and online transactions continue to grow, the number of daily transactions has increased exponentially. Monitoring each transaction manually is practically impossible due to the large volume and complexity of financial data.

Several challenges exist in detecting money laundering activities:

- Extremely large transaction volumes.
- Highly imbalanced datasets where suspicious transactions represent only a small fraction of total transactions.
- Complex transaction behaviors that continuously evolve.
- High false-positive rates generated by traditional rule-based systems.
- Difficulty identifying hidden relationships between transaction attributes.

The dataset used in this project contains millions of legitimate transactions and only a small number of laundering transactions, making accurate classification especially challenging. The objective of this project is to build a Machine Learning model capable of automatically identifying suspicious transactions with high precision and recall while minimizing false alarms. The solution should also support real-time prediction so that potentially fraudulent transactions can be flagged immediately for further investigation.

3. Proposed Solution

To address the challenges of money laundering detection, a Machine Learning-based classification framework was developed. The proposed solution consists of a structured pipeline that transforms raw transaction data into actionable predictions.

The implementation begins with data preprocessing, where transaction records are cleaned and transformed into a format suitable for Machine Learning. Date and time information are merged into a timestamp feature, while categorical variables such as payment currency, bank locations, and payment types are converted into numerical representations using label encoding.

Since the dataset is highly imbalanced, a combination of undersampling and SMOTE (Synthetic Minority Oversampling Technique) is applied. Undersampling reduces the dominance of normal transactions, while SMOTE generates synthetic samples for laundering transactions to create a balanced training dataset.

After preprocessing, multiple classification algorithms are trained and compared, including:

- Logistic Regression
- Decision Tree Classifier
- Random Forest Classifier

Model performance is evaluated using Accuracy, Precision, Recall, and F1-Score. Feature importance analysis is also performed to identify the most influential factors contributing to money laundering detection.

The trained model is then deployed using FastAPI, allowing users to submit transaction details through an API endpoint and receive real-time predictions. The system outputs:

- Prediction (0 = Normal Transaction, 1 = Money Laundering)
- Probability Score
- Detection Status

This solution provides an automated, scalable, and accurate approach for identifying suspicious financial transactions and supporting anti-money laundering efforts within financial institutions.

4. Methodology

The methodology adopted in this project follows a systematic Machine Learning workflow consisting of data preprocessing, feature engineering, class balancing, model training, evaluation, and deployment.

Phase 1: Data Collection

The dataset contains financial transaction records with attributes such as transaction amount, payment currency, sender and receiver bank locations, payment type, laundering category, and transaction timestamp. The target variable, **Is_laundering**, identifies whether a transaction is legitimate or associated with money laundering activities.

Phase 2: Data Preprocessing

The raw transaction data was cleaned and prepared for model development. Data types were analyzed, missing values were handled, and inconsistencies in the dataset were removed.

Since machine learning algorithms require numerical input, categorical attributes such as payment currency, bank locations, payment type, and laundering category were transformed into numerical representations using **Label Encoding**.

Phase 3: Feature Engineering

The separate **Date** and **Time** columns were merged into a single **Timestamp** feature to capture complete transaction timing information.

Additional time-based attributes were extracted from the timestamp, including:

- Year
- Month
- Day
- Hour

These engineered features help the model identify temporal transaction patterns that may indicate suspicious behavior.

Phase 4: Feature Selection

Relevant transaction attributes were selected as input features for machine learning models.

The final feature set included:

- Amount
- Payment Currency
- Received Currency
- Sender Bank Location
- Receiver Bank Location
- Payment Type
- Laundering Type
- Timestamp-Based Features

Feature importance analysis was later performed to identify the most influential variables contributing to laundering detection.

Phase 5: Handling Class Imbalance

The dataset exhibited a significant imbalance between normal and suspicious transactions.

Original Distribution

- Normal Transactions: 9,066,024
- Money Laundering Transactions: 8,980

To overcome this challenge:

Undersampling was applied to reduce the number of majority-class records.

SMOTE (Synthetic Minority Oversampling Technique) was then used to generate synthetic examples of money laundering transactions.

This combination created a balanced training dataset and improved the model's ability to detect rare laundering cases.

Phase 6: Model Development

Three supervised machine learning algorithms were trained and evaluated:

Logistic Regression

A baseline classification model used to establish initial performance benchmarks.

Decision Tree Classifier

A tree-based algorithm capable of learning complex decision rules from transaction data.

Random Forest Classifier

An ensemble learning technique that combines multiple decision trees to improve prediction accuracy and reduce overfitting.

Phase 7: Model Evaluation

The models were evaluated using the following classification metrics:

- Accuracy
- Precision
- Recall
- F1 Score

These metrics were selected because money laundering detection requires not only high accuracy but also the ability to correctly identify suspicious transactions with minimal false negatives.

Phase 8: Model Deployment

The best-performing model was deployed using **FastAPI**.

The deployed API accepts transaction details as input and returns:

- Prediction
- Probability Score
- Detection Status

This enables real-time transaction monitoring and supports automated anti-money laundering investigations.

5. Conclusion

This project successfully developed a Machine Learning-based Money Laundering Detection System capable of identifying suspicious financial transactions with high accuracy. Through comprehensive preprocessing, feature engineering, categorical encoding, and class balancing using SMOTE, the system effectively learned patterns associated with money laundering activities.

Three classification algorithms were evaluated: Logistic Regression, Decision Tree, and Random Forest. Among these models, the **Random Forest Classifier** demonstrated the best performance, achieving:

- **Accuracy:** 99.64%
- **Precision:** 98.57%
- **Recall:** 99.89%
- **F1 Score:** 99.23%

The high recall value indicates that the model successfully identifies the majority of laundering transactions, while the high precision score minimizes false alerts. These characteristics make the model highly suitable for real-world financial crime detection systems.

The trained model was further deployed through a FastAPI-based prediction service, enabling real-time transaction analysis and automated decision-making. The deployment demonstrates the practical applicability of machine learning in enhancing Anti-Money Laundering (AML) operations.

Overall, the project highlights the effectiveness of machine learning techniques in detecting fraudulent financial activities and provides a scalable framework that can be integrated into banking, fintech, and regulatory monitoring systems.

Future Enhancements

- Real-time transaction stream processing.
- Deep Learning-based fraud detection models.
- Graph-based transaction network analysis.
- Explainable AI (XAI) for regulatory compliance.
- Cloud deployment and scalable monitoring infrastructure.
- Advanced anomaly detection techniques for emerging financial crime patterns.